AI STORY & LIVING NOVEL COMPANION

Comprehensive Catalog of Implemented Features, Architecture & Mechanics

Executive Summary: This document provides an exhaustive, detailed breakdown of all implemented features and functional systems within the **AI-Story** ecosystem. The application fuses multi-branching interactive romance and novel writing with an omniscient, deeply living AI companion architecture. Characters possess long-term memory, real-world temporal and meteorological awareness, autonomous peer-to-peer social dynamics, and tactile epistolary artifacts (handwritten diaries, wax-sealed mail, and shared dreamscapes).

1. Living World & Real-Time Reality Synchronization Engine

A foundational architectural pillar designed to dissolve the boundary between fiction and reality by keeping characters connected to the user's physical surroundings in real-time.

• 100% Real-World Weather & Atmospheric Sync

Mechanism: The app senses the user's actual location (city, country, and GPS coordinates), querying live meteorology API data for temperature, cloud cover, and precipitation (Rain, Snow, Thunderstorm, Fog, Clear skies).

Visual Overlays: Custom procedural particle canvas shaders render dynamic rain trickles sliding down the screen glass, soft drifting snow flurries, or twilight haze across the reading and chat interfaces.

Ambient Soundscape Generator: Real-time procedural audio engine blending multi-track soundscapes (window rain, crackling fireplace, midnight cicadas, cozy lo-fi cafe chatter) with independent volume sliders.

Conversational Awareness: Characters naturally acknowledge the weather in 1-on-1 and group chats (e.g. *'Is it pouring where you are too? Stay warm... wish I could share an umbrella with you right now.'*).

• 24-Hour Circadian Routine & Local Time Awareness

Circadian Intelligence: The engine synchronizes with the user's real-time clock (Morning, Midday Lunch, Twilight, Midnight 11 PM–3 AM).

Dynamic Greetings & Tone: Characters alter their dialogue cadence according to the hour (whispering soft sleep-talk late at night, asking if you've eaten lunch at midday, or bringing morning coffee warmth).

2. Autonomous Gossip Drama & Secret Locked Diary

• Living World Autonomous Gossip & Character-to-Character Whispers

Autonomous Continuity: When the user closes the app, characters don't freeze in time; they live their lives and converse with each other behind the scenes about the user's recent choices.

'Overheard Hallway Whispers' Feed: Accessible in Group Chat and Story Details. Displays eavesdropped exchanges with in-world location tags (*Campus Courtyard, Library Stacks, Late-Night Diner*), topic tags (*Jealousy, Teasing, Romance*), and multi-turn dialogue.

'Listen In Now' Eavesdrop Generator: Allows real-time generation of spontaneous character banter based on active chapter incidents.

Chat Reflection: Returning to 1-on-1 chat causes characters to blush or refer back to what was discussed in the gossip feed.

• Secret Locked Diary & Her Private Inner Thoughts

Antique Leather Journal Interface: Styled with aged parchment textures, vintage bookmarks, and brass lock latches.

Hesitation Strikethrough Engine: Parses crossed-out text (e.g., ~I almost held his hand today~) to reveal genuine emotional second thoughts before the character rewrites to play it cool.

Intimacy Progression Lock: Entries are locked behind intimacy thresholds (Level 25, 50, 75, 100) or lock-picking mechanics, rewarding deepened relationships with private emotional confessions.

3. Tactile Romantic Artifacts & Intimacy Sanctuary

A specialized suite of tangible, physicalized in-universe keepsakes and shared activities deepening emotional attachment.

• Vintage Mailbox & Wax-Sealed Letters

Physical Correspondence: Romantic epistolary correspondence featuring customizable wax seals (Crimson, Royal Gold, Midnight Blue) and parchment textures (Vintage Cream, Aged Parchment, Rose Petal).

Clipped Keepsakes & Polaroids: Letters arrive with clipped physical keepsakes (pressed jasmine flowers, dried leaves, silk ribbons) and polaroid snapshots with handwritten captions.

Letter Replies: Users can pen handwritten wax-sealed return letters using custom crests and stationery.

• Shared Dreamworld (Midnight Subconscious Realms)

Surreal Dreamscapes: Unlocked late at night, enabling characters and the user to enter shared subconscious dream environments (*The Starfall Ocean, Floating Amber Lanterns, Forgotten Rooftops*).

Morning-After Continuity: Unlocks secret subconscious confessions that seamlessly transition into morning-after dialogue checks in 1-on-1 chats.

• Domestic Bliss (Cozy Everyday Routine)

Slice-of-Life Tenderness: Simulates quiet domestic intimacy—making morning tea, sharing blanket snuggles, wiping flour off noses while cooking, and gentle playful teasing.

• Candid Camera Roll & Mirror Selfies

Private Gallery: An in-world photo album storing spontaneous candid photos, mirror selfies, and outfit previews generated in real-time.

• VHS Camcorder Vault

Retro Nostalgia: Renders cinematic memories through a nostalgic 90s VHS camcorder aesthetic with scanlines, date-time OSD stamps, and tape grain.

• Romance Boutique & Gifting

Sentimental Tokens: Send curated boutique gifts (chocolates, silver charms, scented mist, ribbon necklaces) prompting dynamic unboxing dialogues and intimacy boosts.

• 4-Cut Photobooth Strip Studio

Retro Booth Strips: Generates Korean/Japanese style 4-cut vertical photobooth strips with stickers, handwritten frames, and cute character expressions.

• Sanctuary Room Inspection

Interactive Exploration: Inspect her personal room, study desk, bedside drawer, and keepsake shelves to discover hidden character lore and backstory notes.

• Midnight Voicemails & Audio Waves

Audio Intimacy: Simulates spontaneous late-night voice messages left when you were unavailable, accompanied by interactive waveform audio visualization.

4. Deep Memory Engine & Conversational Intelligence

• Persistent Long-Term Memory & Vector Semantic Retrieval

ChromaDB / Vector Storage: Encodes user confessions, promises, dates, and emotional milestones into vector embeddings, dynamically recalling relevant past facts into active context.

Chapter Incident Chronicle: Automatically summarizes novel chapter milestones into canon memory so characters never contradict past events.

• Intimacy Progression & Psychological Sentiment Engine

Dynamic Affection Tiers: 0 to 100 Intimacy progression across 4 tiers: *Neutral & Guarded* (0-25), *Warm & Friendly* (26-50), *Romantic & Flirtatious* (51-75), and *Devoted / Deeply Attached* (76-100).

Autonomous Affection Tags: Every conversational turn appends an AI delta tag ([INTIMACY:+1]) that updates character affection in the database in real-time.

Memory Vault Analytics: Displays an emotional breakdown of her feelings, vulnerability levels, attachment style, and relationship status.

• Natural Conversational Cadence & Dialogue Rules

Anti-Monologue Cadence: Enforces realistic texting habits (1 to 3 punchy sentences, 25-50 words max) rather than essay paragraphs.

Mandatory Spoken Dialogue: Ensures at least 70% spoken conversation outside asterisks, restricting action tags to single micro-expressions.

Frequency & Presence Penalties: Configured to eliminate repetitive loops and phrase stagnation.

5. AI Novel Chapter Studio & Copilot Suite

A full-fledged authoring and reading environment allowing seamless generation, inspection, and refinement of serialized story chapters.

• Omniscient 5 Synced Chapter Studio

Synchronized Multi-Chapter Planning: Plans and generates 5 interconnected narrative chapters simultaneously, maintaining overarching plot momentum, character growth arcs, dramatic tension, and romantic climaxes.

Story Arc Mapping: Visualizes pacing, themes, and narrative choices across all 5 chapter beats.

• Story & Character Continuity Checker

Deep Consistency Validation: Scans chapters against character personality profiles, relationship lore, and previous incidents to detect plot inconsistencies, character tone drift, or chronological errors.

Actionable Diagnostic Reports: Provides one-click fix recommendations and continuity scorecards.

• AI Copilot Writer & Editor

One-Click Prose Transformations: Real-time tools for *Rewrite (Enhanced Flow)*, *Dramatize (Heightened Emotion & Stakes)*, *Expand (Sensory & Emotional Depth)*, and *Suggest Next Plot Beats*.

Manga / Comic Panel Visualizer: Transforms narrative text excerpts into structured visual scene prompts and panel dialogue.

Dual Reading Modes & TTS: Reader supports distraction-free reading, customizable fonts, dark/light themes, and built-in Text-to-Speech audio narration.

6. Multi-Character Group Chat & Autonomous Tavern Lounge

Multi-Agent Social Dynamics: Group chat sessions supporting 2 or more characters simultaneously interacting with each other and the user.

Autonomous Turn-Taking: Characters banter, tease, interrupt, defend, and bounce reactions off one another with unique personality cadences.

Shared Novel Continuity: Group conversations maintain omniscient awareness of events and relationships established in the main novel chapters.

7. Technical Architecture & System Specifications

Component	Technology / Implementation Details
Mobile Frontend	Flutter 3.x (Dart), Material Design 3, custom Canvas painters, Google Fonts, GoRouter.
Backend API Server	FastAPI (Python 3.14), asynchronous ASGI architecture, Uvicorn server.
Database & ORM	SQLAlchemy 2.0 Async ORM with SQLite (dev) / PostgreSQL (prod), Pydantic v2 schemas.
Vector Memory	ChromaDB vector embedding store for semantic episodic recall and long-term memory retrieval.
LLM Inference Layer	OpenAI-compatible client integration supporting Gemini, Claude, and local open-weights LLMs.
Audio & Media Engine	audioplayers for procedural multi-track soundscapes and voicemails, gal for image saving.
Build & Deployment	Release APK compilation (AI-Story-Release.apk), automated Git CI/CD to GitHub.